Finals: MA 5104, Department of Mathematics, IIT Madras

Name : Statistical Inference and Learning **Maximum Marks: 100**

- Write **ALL Formulas/Theorems/Results** used.
- Calculators are allowed, but sharing them is NOT allowed.
- You do not need to return the question paper. Only the answer sheet will be collected.
- Write the answers either as a fraction or in decimals up to 3 decimal places.
- Malpractice can result in disciplinary actions as per the Senate mandate.

1. (**24 marks**) Write down ONE correct answer for each of the sub-parts:

(i) Two estimators of θ are available. $\hat{\theta}_A$ is unbiased with variance 9; $\hat{\theta}_B$ has bias $+2$ and variance 4. Which is preferable under mean squared error?

(A) $\hat{\theta}_A$, because unbiasedness is the stronger property.

(B) $\hat{\theta}_A$, because a biased estimator should never be used.

(C) $\hat{\theta}_B$, because its MSE is 8 against 9.

(D) Neither since MSE cannot be compared when one estimator is biased.

(ii) What does the word "confidence" mean in reference to a 95% confidence interval for a population mean μ?

(A) The probability that μ lies in this particular interval is 0.95.

(B) 95% of the sample observations fall inside the interval.

(C) We are 95% confident that the sample mean lies inside the interval.

(D) The procedure produces intervals containing μ in 95% of repeated samples.

(iii) A water board tests every batch of treated water before release, with H_0: the batch is safe. Releasing a contaminated batch is

(A) a type II error.

(B) a type I error.

(C) a correct decision.

(D) impossible to classify without knowing α.

(iv) The bootstrap distribution of an estimator $\hat{\theta}$ is used to approximate

(A) the distribution of the population.

(B) the null distribution of a test statistic for θ.

(C) the posterior distribution of θ.

(D) the sampling distribution of $\hat{\theta}$.

(v) For the same data, a 95% credible interval and a 95% confidence interval come out numerically almost identical. It follows that

(A) the two concepts are equivalent.

(B) they still make different assertions, one about θ, the other about the procedure.

(C) the prior must have been informative.

(D) the sample size must be small.

(vi) The main reason to prefer a non-parametric test to the one-sample t-test is that

(A) it has higher power in all situations.

(B) it does not require the observations to be independent.

(C) it does not require the population to be normal.

(D) it always produces a smaller p-value.

(vii) In a p-dimensional unit cube, a hypercube neighbourhood capturing a fraction r of the data needs edge length $r^{1/p}$. The consequence for kNN in high dimension is that

(A) the neighbours are no longer near, so the bias is large whatever k is chosen.

(B) the variance of the prediction grows without bound.

(C) the computation becomes infeasible.

(D) more observations always repair the problem.

(viii) An analyst screens 500 predictors, keeps the 10 most correlated with y, and then cross-validates a model fitted on those 10. The reported error is

(A) unbiased, since cross-validation was used.

(B) too small, because the screening used the held-out observations.

(C) too large, because only 10 of the 500 predictors were kept.

(D) unaffected, since the screening used only correlations.

2. (**20 marks**) Answer each of the following in **two or three lines**. No credit is given for length.

(i) A courier firm models delivery time (minutes) on distance (km) and vehicle type, reference = motorbike:

$$\widehat{\text{time}} = 12 + 1.8\,\text{dist} + 6.0 \cdot \mathbf{1}\{\text{van}\} - 0.5\,\text{dist} \cdot \mathbf{1}\{\text{van}\}$$

An operations manager reads this as "vans take 6 minutes longer." State what 6.0 actually measures, and give the correct van-minus-bike difference for a 20 km delivery.

(ii) A soil-type predictor has three levels: clay, loam, sand. Explain why it en-

ters the design matrix as two columns rather than three, and state precisely what goes wrong if you use three.

(iii) A utility models monthly electricity consumption (units) on floor area (sq ft) and whether the home has an air conditioner, reference = no air conditioner:

$$\widehat{\text{consumption}} = 85 + 0.21\,\text{area} + 140 \cdot \mathbf{1}\{\text{AC}\}$$

i. State precisely what the coefficients 0.21 and 140 measure.

ii. Which is predicted to consume more, 900 sq ft home with an air conditioner, or a 1200 sq ft home without one? By how much?

(iv) In a dataset of shipments, one predictor is parcel weight. The analyst re-records it in kilograms instead of grams and refits the same least squares model. State what happens to that predictor's coefficient, and what happens to the fitted values and residuals.

3. **(10 (= 3 + 4 + 3) marks)** A researcher has 12 candidate predictors and fits every nested model from 1 predictor up to all 12, recording the training residual sum of squares each time.

(a) Sketch the training residuals sum of squares (RSS) against the number of predictors included, with the number of predictors on the horizontal axis and the training RSS on the vertical axis. State the shape of the curve and explain why it must take that shape.

(b) The researcher concludes that the 12-predictor model is the best of the twelve. Explain what is wrong with this conclusion, and name the quantity that should have been compared instead.

RSS

(c) The same researcher separately fits k nearest neighbours and selects k by minimising the training error. State the value of k that will be chosen and why, and say what this has in common with your answer to part (b).

4. **(10 (=4 + 4 + 2) marks)** Let X and Y be independent, each normally distributed with mean θ, where $\mathrm{Var}(X) = 4$ and $\mathrm{Var}(Y) = 12$. Consider the family of estimators

$$\hat{\theta}_a = aX + (1-a)Y, \qquad 0 \le a \le 1.$$

(a) Show that $\hat{\theta}_a$ is unbiased for every a, and write its MSE as a function of a.

(b) Find the value of a that minimises the MSE, and the resulting MSE.

(c) Comment on how the optimal weights relate to the two variances.

5. **(15 (=5+10) marks)** Let $X_1, \ldots, X_n$ be a random sample from the density

$$f(x;\theta) = \theta x^{\theta-1}, \qquad 0 < x < 1, \quad \theta > 0.$$

You may use the fact that $E(X) = \dfrac{\theta}{\theta+1}$.

(a) Find the method-of-moments estimator of θ.

(b) Find the maximum likelihood estimator of θ.

6. **(15 (= 3*5) marks)** A chemical process is run with a new catalyst and with the existing one, and the yield (%) is recorded. Two independent random samples from normal populations give

	n	$\bar{x}$	s^2
New catalyst	13	64.3	16.8
Existing catalyst	10	58.9	3.5

(2.56099, 8.23901)

(a) Assuming $\sigma_1^2 = \sigma_2^2$, find a 95% confidence interval for $\mu_1 - \mu_2$. Note $t_{0.025,21} = 2.080$. Show the pooled variance explicitly. 3.3245

(b) Interpret the interval, and state what you would tell the plant manager about the new catalyst. new

(c) Note $F_{0.05,12,9} = 3.07$ and $F_{0.05,9,12} = 2.80$. Construct a 90% confidence interval for σ_1^2/σ_2^2. $\left(\frac{480}{307}, \frac{336}{25}\right)$

(d) In the light of (c), was the assumption made in part (a) justified? State clearly what you conclude and what you would do differently. $\sigma_1^2 \neq \sigma_2^2$

(e) One of your classmates argues that because the assumption fails, the interval in (a) is too narrow and the conclusion in (b) cannot be trusted. Is the classmate right? Justify your answer by reference to which sample carries the larger variance. he is wrong, because a has higher width

7. **(6 marks)** A quality characteristic is normally distributed with known standard deviation $\sigma = 8$. A sample of $n = 25$ is used to test $H_0 : \mu = 100$ against $H_1 : \mu \neq 100$, and H_0 is rejected if $|\bar{X} - 100| > 4$.

(a) Find the probability of committing a type II error when $\mu = 104$ in the alternative hypothesis. $\Phi(0) - \Phi(-1)$

55/27

4/22 5.7

4/24 4.8

307/480